EE 122: Introduction to Control Systems

Problem Set #11: Loop shaping-based controller design

Name: ________________ **Submission Date:** ________________

Problem 1 Consider the second-order plant

$$G(s) = \frac{1}{s^2 + 6s + 5}.$$

In Figure 1, you are given the desired loop gain magnitude plot for

$$L(s) = C(s)G(s).$$

Your goal is to design a compensator using loop-shaping ideas so that the loop transfer function approximately matches the desired magnitude plot.

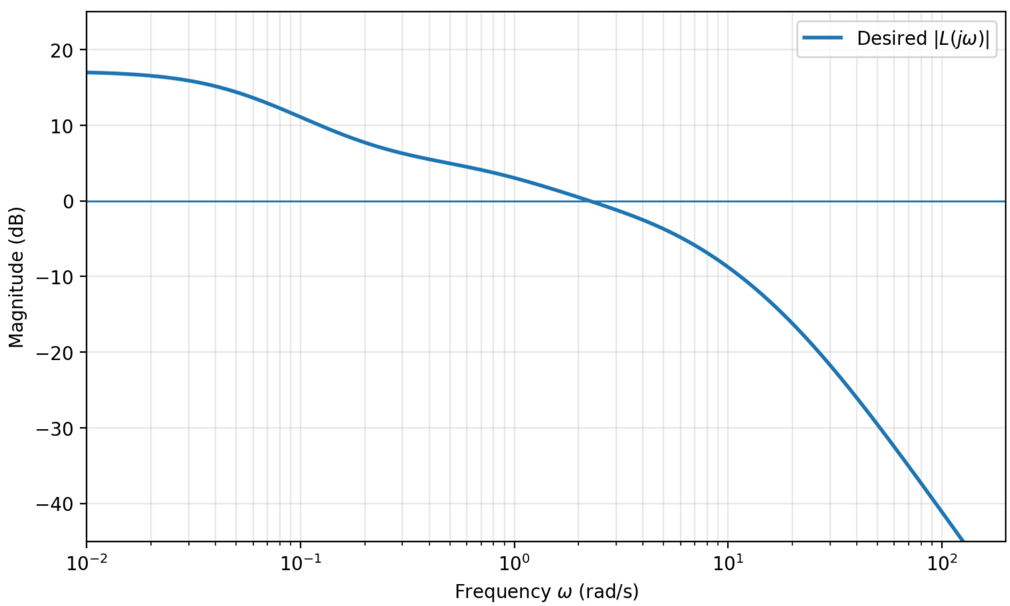

Figure 1: Desired loop-gain magnitude plot.

(a) [**10 points**] Start from the plant $G(s)$ and sketch its asymptotic Bode magnitude plot.

(b) [**30 points**] From the desired loop-gain plot, determine how many additional poles and zeros must be contributed by the compensator. Write down the structure of the lead-lag compensator $C(s)$ that you will use for this design.

(c) [**10 points**] Estimate the break frequencies of the compensator by comparing the slope changes in the desired plot against the known break frequencies of the plant.

(d) [**10 points**] Identify which part of your compensator acts as the *lag* portion and which part acts as the *lead* portion.

(e) [**10 points**] Estimate the gain K needed so that the compensated loop transfer function

$$L(s) = C(s)G(s)$$

matches the desired magnitude plot.

(f) [**10 points**] Write your final compensator $C(s)$ explicitly.

(g) [**20 points**] Plot the Bode magnitude of your designed loop transfer function and compare it against the desired plot.